It is recommended that you spend about 30 minutes on this question.

2 One type of radiation detector known as a Geiger-Müller tube is shown in Fig. 2.1.

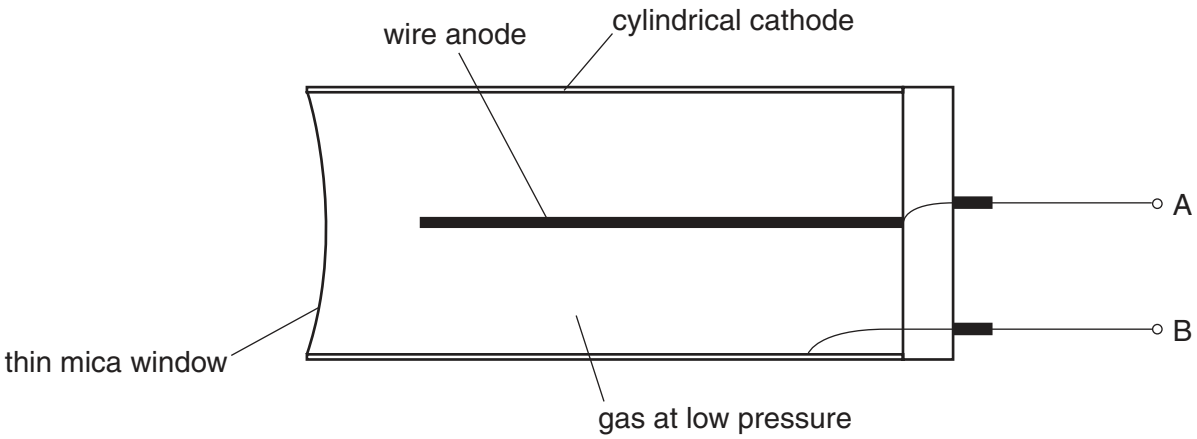

Fig. 2.1

In order for the tube to function, a potential difference V_{AB} has to be applied between A and B. The count rate registered by an instrument connected to the tube depends on several factors such as the distance from the radioactive source to the thin mica window, the activity of the source and V_{AB}. The tube can detect α, β and γ-radiation.

Design a laboratory experiment to investigate how the count rate due to γ-radiation **only** depends upon the potential difference V_{AB}. You have access to three different radioactive sources only. Information relating to each of these sources is given in the table below.

source	type of radiation emitted	half-life of source
Radium-226	α, β and γ	1600 years
Bismuth-214	β and γ only	20 minutes
Cobalt-60	β and γ only	5 years

You may assume that the following equipment is available, together with any other apparatus that may be found in a school or college science laboratory.

Aluminium plates of different thicknesses
Ammeter
Connecting wires
Datalogger
Geiger-Müller tube
Lead plates of different thicknesses
Metre rule
Oscilloscope
Ratemeter
Scaler
Signal generator
Source handling tool
Variable d.c. power supply
Voltmeter

[Turn over

https://xtremepape.rs/

You should draw a diagram showing the arrangement of your apparatus. In your account you should pay particular attention to

(a) which source you would use, giving a reason for your choice,

(b) the procedure to be followed, including how the count rate would be measured,

(c) the control of variables,

(d) any safety precautions you would take.

.....

.....

.....

.....

.....

.....

.....

.....

.....

https://xtremepape.rs/